

RIFT: Rule-Induced Framework for Single-Source Cross-Dataset Generalization in Rumor Detection

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Abstract

We study single-source cross-dataset generalization in rumor detection, where the detection model trained in the source dataset should be tested in the target dataset without training. This challenges the model to directly generalize under the situation that the target dataset can have significant differences from the source in major aspects such as topics of information and cascades of spreading, while existing rumor detection approaches usually lack such strong generalizability. Large Language Models (LLMs) are promising in this task due to their general understanding and reasoning abilities, while they remain struggling due to failure in grasping key information and conducting consistent and robust reasoning. To tackle these challenges, we propose *RIFT*, a novel framework to enhance LLM-based rumor detection, whose key idea lies in utilizing LLMs to induce a set of logical rules from the source to serve as explainable and consistent predictive evidence. To sufficiently distill source dataset information into transferrable logical knowledge, we propose a GCN-guided rule induction pipeline that converts source textual, conversational, and propagation signals encoded by Graph Convolution Networks (GCNs) into logical rules for LLM-based target dataset inference. Experiments in challenging rumor detection scenarios show that, by inducing explainable and consistent logical rules from data, our approach significantly improves the performance of LLMs in cross-dataset generalization. Our code is released at (link available upon acceptance).

1 Introduction

Social media has become a major channel for discussing emerging events, but it also enables rumors to spread rapidly across users and communities. Since delayed correction may not fully eliminate the influence of widely propagated rumors, practical rumor detection systems need to make early judgments from limited information, such as the

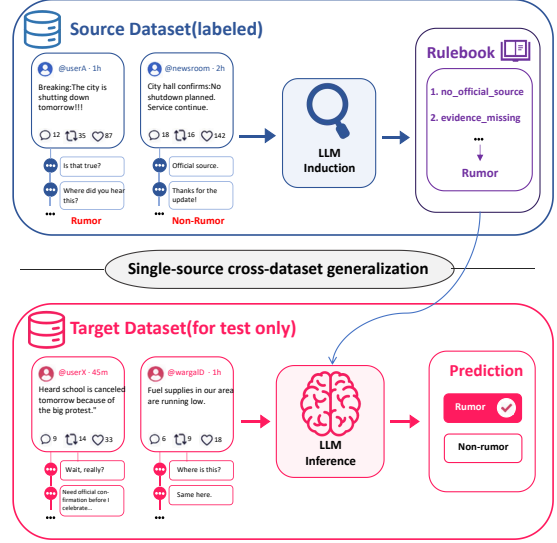


Figure 1: The rule-induced framework for single-source cross-dataset generalization in rumor detection.

source post, early replies, and initial propagation signals. Following these practical considerations, we study a challenging real-world setting for rumor detection: single-source cross-dataset generalization, where the detection model trained in the source dataset should be tested in the target dataset without training. In reality, rumors may emerge around different events, topics, platforms, and user communities, while collecting labeled data for every new situation is costly and slow. Models trained on one source dataset may, therefore, fail in a target dataset due to shifts in language style, event background, user behavior, label distribution, and propagation patterns. This makes cross-dataset generalization an important but difficult setting.

Although rumor detection has been extensively studied in the literature, few existing works tackle cross-dataset generalization. Supervised propagation- and graph-based rumor detectors (Sun et al., 2022; Bian et al., 2020) have already achieved remarkable in-dataset performance, while

they lack generalizability across datasets. Cross-domain rumor or misinformation detection methods further significantly improve transferability through adversarial learning (Ma et al., 2021; Chen et al., 2025), contrastive learning (Ran and Jia, 2023), domain alignment (Gong et al., 2025), mixture-of-experts (Qin et al., 2020; Liu et al., 2025a), or meta-learning (He et al., 2025). While most approaches still require further adaptation under the target dataset, making their direct cross-dataset generalization ability limited.

On the other hand, due to general understanding and reasoning abilities, Large Language Models (LLMs) have strong potentials in this task. However, we observe that the power of LLMs cannot be properly elicited when asked to predict directly without any adaptation, due to their potential failure to grasp key information and conduct consistent and robust reasoning. This leads to failure under the cross-dataset generalization setting, in which the adaptation under the target dataset is invalid. Furthermore, this raises a fundamental technical question: To elicit the generalizability of LLMs, what kind of knowledge is effective to transfer from the source to the target dataset?

In this paper, we explore this question by proposing *RIFT*, a rule-level transfer framework that distills source-dataset rumor patterns into induced symbolic rules and uses them to guide LLM-based target-dataset inference. Recent studies have justified that symbolic rules can serve as composite and generalizable knowledge for LLMs in solving challenging reasoning tasks (Gao et al., 2023; Li et al., 2023; Chae et al., 2024; Ling et al., 2023; Peng et al., 2025). In particular, it has been shown that such symbolic rules can be autonomously induced by LLMs from data (Piriyakulkij et al., 2025; Jones et al., 2025; Hsu et al., 2025). This paradigm is promising in principle for cross-dataset generalization due to its advantages in improving the faithfulness, interpretability, and generalizability of LLMs with symbolic knowledge. While for rumor detection, it remains challenging to extract representative and essential rules from complicated topics of information and cascades of spreading.

We tackle this challenge by proposing a rule induction pipeline guided by Graph Convolutional Networks (GCNs). Specifically, GCN-based detection models, which are justified for their strong in-domain performance, are introduced to identify highly informative samples for induction. Subsequently, a semantic grounding approach is pro-

posed to extract their textual, conversational, and propagation-related signals into Boolean predicates. Finally, we propose a rule induction method by which candidate rules are automatically induced and scored before being injected into the LLM prompt as natural-language guidance (Qiao et al., 2021; Pryzant et al., 2022; Wei et al., 2022; Zhu et al., 2023). Experiments in challenging rumor detection scenarios show that, by inducing explainable and consistent logical rules from data, our approach significantly improves the performance of LLMs in cross-dataset generalization.

2 Related Work

2.1 Supervised Rumor Detection with Propagation Signals

Early rumor detection studies mainly formulated the task as supervised classification using textual, temporal, user-level, and propagation-based features (Qazvinian et al., 2011) (Castillo et al., 2011). Unlike generic text classification, rumor detection often relies on signals beyond the source post itself, since propagation patterns and user reactions can provide important cues for veracity assessment. Early propagation-aware methods therefore modeled temporal evolution and diffusion structures to improve rumor detection performance (Ma et al., 2015, 2016, 2017).

Later neural approaches reduced manual feature engineering by learning representations from posts, replies, and propagation paths. Tree-structured neural models encoded reply structures along propagation trees (Ma et al., 2018b; Cui and Jia, 2024), while graph neural networks represented rumor propagation as graph-structured data. For example, BiGCN models both top-down diffusion and bottom-up response patterns (Bian et al., 2020), and later graph-based methods further improved in-domain rumor detection by exploiting propagation dependencies (Sun et al., 2022; Huang et al., 2022). Other supervised approaches also incorporate stance, conversation context, or explanatory signals from user comments (Kochkina et al., 2018; Ma et al., 2018a; Shu et al., 2019).

Although these methods achieve strong in-domain performance, they often rely on domain-specific propagation patterns and interaction behaviors, making the learned representations difficult to generalize across domains. Moreover, the transferred knowledge is usually encoded implicitly in neural parameters, making it difficult to inspect

which source-domain patterns remain reliable after distribution shifts. Instead of directly transferring graph-model parameters, we use graph models to identify reliable source-domain patterns for later rule induction and cross-domain inference.

2.2 Cross-Domain Misinformation Detection

Cross-domain misinformation detection studies how to transfer detection ability from labeled source domains to unseen or weakly labeled target domains. This setting is challenging because topics, events, platforms, user behaviors, propagation patterns, and label distributions may differ substantially across domains.

Existing studies improve cross-domain generalization through several strategies. Multi-domain and mixture-of-experts methods model domain-specific and domain-shared patterns through expert routing mechanisms (Qin et al., 2020) (Nan et al., 2021), while shared-private representation learning methods disentangle transferable and domain-specific features (Silva et al., 2021). Domain adaptation approaches further align source and target distributions through adversarial learning, contrastive learning, pseudo-labeling, or distribution alignment (Ran and Jia, 2023; Ran et al., 2026; Gong et al., 2025; Chen et al., 2025). Meta-learning (Yue et al., 2023) and few-shot adaptation (Alghamdi et al., 2024; Jiang et al., 2025) methods instead aim to improve rapid adaptation to unseen domains with limited supervision.

Despite their differences, most existing methods still transfer knowledge through latent representations, adapted parameters, or expert routing weights, making the transferred knowledge difficult to inspect or explicitly reuse during downstream reasoning. In contrast, our work transfers explicit symbolic rules distilled from source-domain patterns, enabling a more inspectable and controllable form of cross-domain knowledge transfer.

2.3 LLM-based and Retrieval-Augmented Misinformation Detection

LLMs have recently been applied to misinformation detection, fact-checking, and rumor verification through zero-shot and few-shot prompting (Cheung and Lam, 2023; Hu et al., 2024; Asadi et al., 2025; Tian et al., 2025; Zeng et al., 2026). Compared with fully supervised models, LLMs provide a flexible inference interface and can incorporate demonstrations or external evidence during reasoning. Retrieval-augmented and in-context

learning methods further improve LLM inference by retrieving relevant examples, evidence, or external knowledge (Li et al., 2024; Singal et al., 2024; Chen et al., 2024; Ge et al., 2025; Liu et al., 2025b). Recent cross-dataset approaches also combine retrieval with LLM prompting. For example, RAE-moLLM retrieves affect-related source-domain examples and uses them as in-context demonstrations for target-domain prediction (Liu et al., 2025b). These methods show that retrieved context can improve cross-domain misinformation detection without full model fine-tuning. Unlike retrieval-augmented methods that directly place retrieved examples or evidence into the final prompt, our framework uses source-domain samples to induce predicates and symbolic rules before target-domain inference. In this framework, retrieval is not the final transferred knowledge unit; selected rules are. This shifts cross-domain transfer from retrieved examples toward explicit rule-level knowledge that is more compact, inspectable, and controllable.

3 Method

This section introduces our RIFT (Rule-Induced Framework for rumor deTectioN) approach, whose pipeline is illustrated in Fig. 1 and 2.

3.1 Task Definition

We focus on single-source cross-dataset rumor detection, where labeled events are available in a source dataset and the model is evaluated on events from a different unlabeled target dataset. We formulate rumor detection as a binary classification task over social media events. Let $\mathcal{D}_s = \{(C_i^s, y_i^s) \mid i = 1, \dots, N_s\}$ denote the labeled source dataset, where C_i^s is the i -th source event and $y_i^s \in \mathcal{Y}$ is its label. The label space is shared across datasets, with $\mathcal{Y} = \{0, 1\}$, where 1 denotes rumor and 0 denotes non-rumor.

Each event consists of an original post and its reply context. Specifically, we represent a source event as $C_i^s = (p_i^s, R_i^s)$, where p_i^s denotes the original post and $R_i^s = \{r_{i,1}^s, r_{i,2}^s, \dots, r_{i,n_i}^s\}$ denotes its observed replies. Similarly, the target dataset is denoted as $\mathcal{D}_t = \{C_j^t \mid j = 1, \dots, N_t\}$, where each target event is represented as $C_j^t = (p_j^t, R_j^t)$. Given $\mathcal{D}_s$, the objective is to predict the label $\hat{y}_j^t \in \mathcal{Y}$ for each target-dataset event C_j^t .

3.2 Source Context Construction Module

This module constructs *source context examples* for subsequent semantic grounding and symbolic

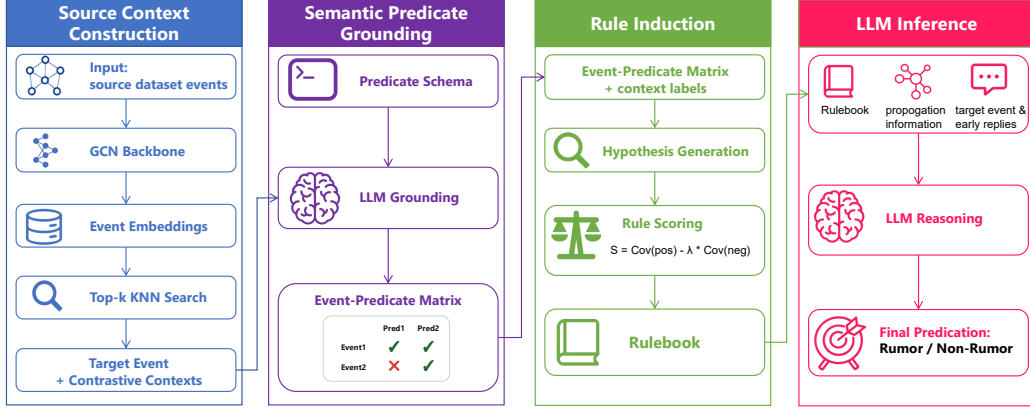


Figure 2: Illustration of the GCN-guided rule induction method in *RIFT*.

rule induction. It takes as input the labeled source dataset $\mathcal{D}_s$ and a graph-based rumor detection backbone, such as BiGCN or DDGCN, trained on $\mathcal{D}_s$. Its output is a set of context examples $\{S_m\}_{m=1}^M$, where each S_m contains a query event and its retrieved rumor/non-rumor neighbors, together with their original posts and early replies.

Generating query events. We first train a graph-based rumor detection backbone on $\mathcal{D}_s$ and extract its final hidden representation $\mathbf{h}_i^s \in \mathbb{R}^d$ for each source event C_i^s . Since graph-based detectors are trained to distinguish rumor and non-rumor events by modeling textual and propagation information, their hidden representations provide task-aware event embeddings. To avoid noisy or ambiguous contexts, we select high-confidence rumor events as queries. Specifically, we rank all source events with ground-truth rumor labels by the backbone’s predicted probability for the rumor class and choose the top M events as query events, denoted as $\mathcal{Q} = \{q_m\}_{m=1}^M$.

Retrieving and formatting contrastive contexts. For each query event q_m , we retrieve its nearest neighbors in the backbone representation space using cosine similarity. To provide contrastive evidence, we separately retrieve the top- k most similar rumor events, excluding q_m itself, and the top- k most similar non-rumor events from the source training set. The resulting neighborhood is $\mathcal{B}_m = \{q_m\} \cup \mathcal{N}_m^R \cup \mathcal{N}_m^{NR}$, where $\mathcal{N}_m^R$ and $\mathcal{N}_m^{NR}$ denote the retrieved rumor and non-rumor neighbor sets, respectively. Including both rumor and non-rumor neighbors encourages the following modules to focus on discriminative evidence rather than merely recurring topic-specific patterns.

We then convert each neighborhood $\mathcal{B}_m$ into a structured context example S_m . For every event in $\mathcal{B}_m$, we include its original post, the earliest N replies in chronological order, and the corresponding reply timestamps. In our implementation, $N = 15$ is used to balance early-context coverage and input length. The query event, rumor neighbors, and non-rumor neighbors are separated with explicit markers. The resulting set $\{S_m\}_{m=1}^M$ is passed to the semantic grounding module.

3.3 Semantic Predicate Grounding Module

This module converts each source-dataset context example into a structured predicate representation. Its input is a source-dataset context S_m constructed in the previous module, and its output is an event-predicate matrix $\mathbf{Z}^{(m)}$ together with the ground-truth rumor labels of the involved events.

We define a fixed predicate schema $\mathcal{P} = \{p_1, p_2, \dots, p_L\}$ that captures dataset-general rumor-related characteristics. These predicates are inspired by commonly studied cues in the rumor detection literature (Castillo et al., 2011; Qazvinian et al., 2011; Zhao et al., 2015; Arif et al., 2017; Zubiaga et al., 2018; Starbird et al., 2018), and include aspects such as source attribution, evidential support, claim specificity, uncertainty or speculation, emotional intensity, and early conversational reactions. The schema is kept identical across source and target datasets because these predicates are designed to reflect *dataset-invariant* linguistic and conversational patterns. The predicate definitions and their natural language descriptions are provided in Appendix A for completeness.

Given a source-dataset context S_m , we use an LLM to ground each event in the context onto

the predicate schema. Specifically, for each event $e_i \in \mathcal{B}_m$ and each predicate $p_l \in \mathcal{P}$, the LLM determines whether the event satisfies the predicate based on its original post, early replies, and reply timestamps. The grounding result is a Boolean value $z_{i,l}^{(m)} \in \{0, 1\}$, where $z_{i,l}^{(m)} = 1$ indicates that event e_i satisfies predicate p_l . The LLM is instructed with a fixed prompt that describes each predicate in natural language; the prompt template is deferred to Appendix B due to space constraints. For each event, the LLM returns a vector of L Boolean values, which we collect into the event-predicate matrix $\mathbf{Z}^{(m)} \in \{0, 1\}^{|\mathcal{B}_m| \times L}$.

Thus, for each context S_m , we obtain a tuple $(\mathbf{Z}^{(m)}, \mathbf{y}^{(m)})$, where $\mathbf{y}^{(m)} \in \{0, 1\}^{|\mathcal{B}_m|}$ contains the ground-truth rumor labels of the events in $\mathcal{B}_m$ provided by the source dataset. The collection $\{(\mathbf{Z}^{(m)}, \mathbf{y}^{(m)})\}_{m=1}^M$ is then passed to the symbolic rule induction module.

3.4 Rule Induction Module

This module induces a compact rumor rulebook from the grounded predicate representations. Its input is the collection of event-predicate matrices and labels $\{(\mathbf{Z}^{(m)}, \mathbf{y}^{(m)})\}_{m=1}^M$, and its output is a textual rulebook $\mathcal{R}^*$ consisting of rumor-indicative symbolic rules.

We define a compact rule language over the predicate schema $\mathcal{P}$. To keep the induced rules interpretable and the search space tractable, we restrict that a candidate rule can be either a single predicate $p_a \rightarrow \text{Rumor}$ or a pairwise conjunction $p_a \wedge p_b \rightarrow \text{Rumor}$, where $p_a, p_b \in \mathcal{P}$.

For each context S_m , we evaluate every candidate rule r on the corresponding matrix $\mathbf{Z}^{(m)}$. A rule is activated for an event only when all predicates in the rule are true for that event. We score each rule by rewarding its coverage of rumor events and penalizing its coverage of non-rumor events:

$$S_m(r) = \text{cov}_R^{(m)}(r) - \lambda * \text{cov}_{NR}^{(m)}(r),$$

where $\text{cov}_R^{(m)}(r)$ and $\text{cov}_{NR}^{(m)}(r)$ denote the proportions of rumor and non-rumor events in $\mathcal{B}_m$ covered by rule r , respectively. The coefficient λ controls the penalty for covering non-rumor events. We keep only rules with $S_m(r) > 0$ as valid candidate rules for the current context.

Some retrieved contexts may be noisy or overly ambiguous, causing all candidate rules to receive non-positive scores. In such cases, we apply a

lightweight resampling strategy: we randomly remove events from the context and repeat the rule search, while requiring at least two remaining events for each class. This constraint prevents rules from being induced from a single idiosyncratic example. For all valid rules collected from different contexts, we compute an aggregate score:

$$A(r) = \sum_{m=1}^M \mathbb{I}[r \in \mathcal{R}_m] \cdot S_m(r),$$

where $\mathcal{R}_m$ is the valid rule set induced from context S_m . The aggregate score favors rules that appear frequently across contexts and receive high local scores. Finally, we select the top- K rules according to $A(r)$ and verbalize them into a textual rumor rulebook $\mathcal{R}^*$, which is transferred to the target-dataset inference stage.

3.5 Rule-enhanced Inference Module

This module predicts the label of each target-dataset event using the induced rumor rulebook. Given a target event $C_j^t = (p_j^t, R_j^t)$ and the learned rulebook $\mathcal{R}^*$, we construct an inference prompt that includes three components: the source-induced rumor rules, the target event content, and a brief propagation summary. The target event content contains the original post and its observed early replies, while the propagation summary describes basic structural signals such as reply volume and temporal response patterns. The LLM is instructed to judge whether the activated rule patterns are consistent with the target event and to make the final prediction based on the overall evidence.

4 Experiments

4.1 Experimental Setup

Datasets. We adopt three widely used rumor detection datasets: PHEME (Kochkina et al., 2018), Twitter16 (Ma et al., 2017), and Twitter-COVID19 (Lin et al., 2022). These datasets cover different events, topics, and time periods, making them suitable for evaluating cross-dataset rumor detection. Following the binary rumor detection setting, we map each instance into one of two labels: rumor and non-rumor. The detailed statistics of the datasets are shown in Table 6 of the appendix.

Baselines. We compare our method with four groups of baselines. First, we include text-based non-LLM classifiers, including BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), which

LLM Backbone	Method	PHEME→Twitter-COVID19				PHEME→Twitter16				Twitter16→Twitter-COVID19			
		Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
non-LLM	BERT	42.00	95.45	8.33	15.33	40.59	84.70	25.29	38.94	45.25	59.02	42.86	49.66
	RoBERTa	53.25	83.51	32.14	46.42	47.07	79.41	39.64	52.88	41.50	55.63	35.32	43.20
	BIGCN	43.50	74.07	15.87	26.14	32.67	71.85	16.09	26.29	40.50	59.72	17.06	26.54
	DDGCN	47.00	91.67	17.46	29.33	40.60	77.50	24.55	37.29	43.25	58.17	35.32	43.95
	PRL	53.06	91.43	28.19	43.09	48.30	80.00	41.30	54.48	38.33	55.56	11.01	18.38
Deepseek-v4-flash	Zero-shot	56.25	72.51	49.21	58.63	51.47	91.86	38.66	54.42	56.25	72.51	49.21	58.63
	RAEmoLLM	53.75	74.45	40.48	52.44	63.35	83.22	57.67	68.13	49.75	62.69	50.00	55.63
	RIFT (Ours)	65.50	77.40	63.89	70.00	66.75	85.30	67.21	75.18	65.50	75.68	66.67	70.89
Qwen3.5-plus	Zero-shot	57.07	78.68	43.15	55.73	57.82	90.36	48.94	63.49	57.07	78.68	43.15	55.73
	RAEmoLLM	54.00	68.48	50.00	57.80	66.98	86.96	60.47	71.33	50.50	62.62	53.17	57.51
	RIFT (Ours)	67.09	76.92	68.00	72.19	70.05	88.90	68.60	77.44	63.25	72.73	66.67	69.57
GPT-4o	Zero-shot	64.50	73.50	68.25	70.78	65.89	84.65	66.56	74.52	64.50	73.50	68.25	70.78
	RAEmoLLM	56.50	66.39	62.70	64.49	72.86	84.60	77.98	81.15	50.25	59.93	63.49	61.66
	RIFT (Ours)	70.75	79.48	72.22	75.68	74.08	78.77	89.56	83.82	75.25	74.60	92.06	82.42

Table 1: Baseline comparisons w.r.t. Accuracy, Precision, Recall, and F1. The best result within each LLM group and transfer direction is highlighted in bold.

perform supervised rumor classification based on the textual content of the source post and replies. Second, we compare with graph-based rumor detectors, including BiGCN (Bian et al., 2020) and DDGCN (Sun et al., 2022), which model social media events as propagation graphs and learn structural representations for rumor detection. These models are trained under the source dataset and directly tested under the target dataset.

Third, we consider cross-domain detection methods. PRL (Lin et al., 2023) is included as a cross-domain rumor detection baseline. We also compare with RAEmoLLM (Liu et al., 2025b), an LLM-based cross-domain misinformation detection method that can be adapted to our setting. Finally, we include zero-shot prompting baselines with different LLMs, including DeepSeek-v4-flash (DeepSeek-AI, 2026), Qwen3.5-plus (Qwen Team, 2026), and GPT-4o (OpenAI, 2024). For each LLM, the zero-shot baseline uses the same target event information as our method but does not use the induced rulebook. This allows us to examine whether the proposed rule-enhanced inference brings consistent gains over direct LLM prediction.

Early detection setting. To simulate realistic cross-dataset scenario, we adopt a challenging evaluation setting where only early conversational information is available. Specifically, for each event, we retain at most the replies posted within the first 30 minutes after the initial post. This restriction prevents models from relying on later discussions

or corrections that may not be available in real-time deployment. Furthermore, when using LLMs for inference, we include at most 15 reply texts for each test event due to context-window limitations and to ensure comparable input budgets across methods. Therefore, each test instance consists of the source post and a limited set of early replies, requiring models to make rumor judgments from partial and early-stage propagation evidence.

4.2 Baseline Comparison

Table 1 reports the baseline comparisons under three cross-dataset rumor detection settings. The non-LLM baselines show limited cross-dataset generalization performance. Text-based models such as BERT and RoBERTa rely mainly on lexical and semantic cues, while graph-based models such as BiGCN and DDGCN directly transfer source-dataset representations to the target dataset. Their low accuracy and recall indicate that supervised rumor detectors are sensitive to distributional shifts in topics, language usage, and propagation patterns.

Zero-shot LLMs generally perform better than non-LLM baselines, showing their stronger general semantic understanding abilities. However, they suffer from low recall frequently. For example, DeepSeek-v4-flash obtains only 38.66% recall on PHEME→Twitter16, and Qwen3.5-plus obtains 43.15% recall on PHEME→Twitter-COVID19 and Twitter16→Twitter-COVID19. This suggests that LLMs tend to make conservative rumor judgments without task-specific guidance. Meanwhile, RAE-

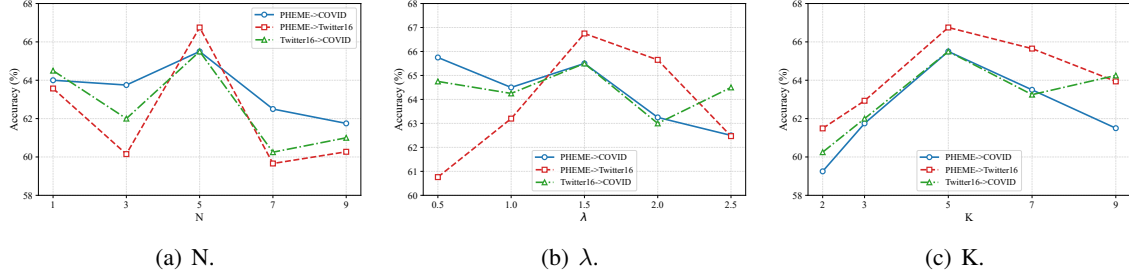


Figure 3: Ablation study results on parameter sensitivity.

Module	Variant	PHEME→ COVID19	PHEME→ Twitter16	Twitter16→ COVID19
Full	Full model	70.75	74.08	75.25
Context	Rand. neigh.	64.00	70.05	71.75
Induction	Rand. rules	66.75	70.29	70.50
Inference	No prop. summ.	68.50	71.88	72.25

Table 2: Accuracy under different ablation settings.

Setting	PHEME→ COVID19	PHEME→ Twitter16	Twitter16→ COVID19
BiGCN	43.50	32.67	40.50
BiGCN + Ours	70.75	74.08	75.25
DDGCN	47.00	40.60	43.25
DDGCN + Ours	70.25	73.59	70.75

Table 3: Accuracy with different GCN backbones.

moLLM is not able to fully address this issue despite improvement in some settings.

In contrast, our method consistently achieves the best accuracy and F1 within each LLM block across all three transfer settings. Compared with zero-shot prompting, our method improves the average accuracy by 11.26 points with DeepSeek-v4-flash, 9.48 points with Qwen3.5-plus, and 8.40 points with GPT-4o. The gains are especially clear in recall: for GPT-4o, our method improves recall from 66.56% to 89.56% on PHEME→Twitter16 and from 68.25% to 92.06% on Twitter16→Twitter-COVID19. These results demonstrate that transferring source-dataset knowledge as symbolic rule evidence helps LLMs identify suspicious or insufficiently supported claims that are often missed by zero-shot prompting.

4.3 Ablation Study

Pipeline modules. We conduct module-level ablation experiments to evaluate the contribution of each component in our pipeline. As shown in Table 2, we compare the full model with three variants: *Random neighbors*, which replaces KNN-based source-dataset retrieval with random neighbor selection; *Random rules*, which randomly selects rules from positive-scoring candidates instead of using our ranking strategy; and *No propagation summary*, which removes target-dataset propagation statistics from the inference prompt.

The full model achieves the highest accuracy across all three transfer settings. Replacing KNN

retrieval with random neighbors consistently reduces performance, indicating that the backbone representation space provides useful task-aware neighborhoods for constructing source-dataset contexts. The *Random rules* variant also leads to clear drops, showing that the proposed rule scoring and aggregation mechanism is important for selecting transferable rumor-indicative rules. Removing the propagation summary results in smaller but consistent performance decreases, suggesting that propagation statistics provide complementary structural evidence during inference.

Parameter sensitivity We further analyze the sensitivity of three key hyperparameters: the rule set size N , the number of retrieved neighbors K , and the non-rumor penalty coefficient λ .

For the rulebook size N , we observe that $N = 5$ achieves the best performance across all three settings. A small rulebook may provide insufficient coverage of diverse rumor-indicative patterns, while adding too many rules can introduce lower-ranked or redundant evidence into the prompt. This suggests that a moderate-size rulebook better balances evidence coverage and rule reliability.

For the neighbor number K , performance also peaks around $K = 5$. When K is too small, the retrieved context may not contain enough contrastive evidence for reliable rule induction. In contrast, larger values of K may introduce less relevant neighbors, which can make the grounded predicates and induced rules noisier. Thus, $K = 5$ pro-

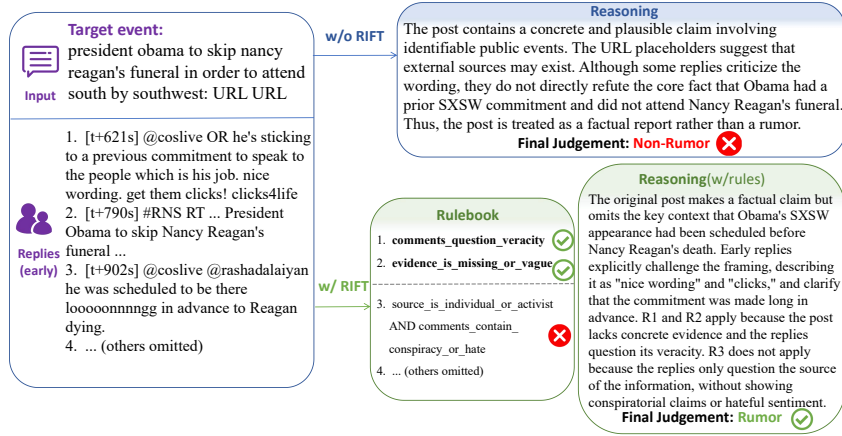


Figure 4: Case study of a rumor correctly classified by RIFT while misclassified without it.

vides a reasonable trade-off between local context richness and retrieval noise.

For the penalty coefficient λ , the results show that an intermediate value generally performs better. A small λ may select overly broad rules that also cover non-rumor events, while a large λ may over-penalize candidate rules and make rule selection too conservative. Overall, the default setting $N = 5$, $K = 5$, and $\lambda = 1.5$ provides a stable balance across different transfer settings.

GCN Backbone Analysis. We analyze whether our framework depends on a specific graph backbone. We use GPT-4o as the inference LLM and report accuracy under three transfer settings. As shown in Table 3, our method consistently improves both BiGCN and DDGCN. With BiGCN, the average accuracy increases from 38.89% to 73.36%, while with DDGCN it increases from 43.61% to 71.53%. This indicates that the proposed framework is not tied to a particular graph encoder, but can benefit from different backbone representations for source-dataset context construction and rule induction.

The direct transfer results of BiGCN and DDGCN are relatively low across the three settings, showing the difficulty of cross-dataset generalization for conventional graph-based detectors. In contrast, our framework substantially improves accuracy in all transfer directions, suggesting that source-dataset knowledge becomes more transferable when abstracted into symbolic rule evidence and used to guide LLM-based inference.

4.4 Case Study

Figure 4 shows a case where zero-shot prompting misclassifies, while our rule-enhanced inference

makes the correct prediction. The target post claims that President Obama skipped Nancy Reagan’s funeral to attend South by Southwest. The zero-shot LLM focuses on the plausibility of the event and the existence of URL placeholders, and therefore treats the post as a factual report. However, it overlooks important early-reply signals, where users question the framing of the claim and point out that Obama’s SXSW appearance had been scheduled before Nancy Reagan’s death.

With our method, the induced rulebook provides additional rumor-specific evidence. Rules such as `comments_question_veracity` and `evidence_is_missing_or_vague` guide the LLM to consider whether the post lacks sufficient evidence and whether early replies challenge the claim. As a result, the rule-enhanced model recognizes the disputed and context-missing nature of the post and correctly predicts *Rumor*.

5 Conclusion

We study single-source cross-dataset generalization in rumor detection, where labeled data are available only from a source dataset and the model must detect rumors in unseen target datasets. We propose RIFT, a rule-enhanced LLM inference framework that transfers source knowledge as interpretable logical rules. We propose a GCN-guided rule induction pipeline to reliably distill source dataset information into transferrable logical knowledge. Experiments across multiple cross-dataset settings show that our method consistently improves over strong LLM and non-LLM baselines, demonstrating that symbolic rumor evidence introduced in our approach provides an effective and efficient way to improve cross-dataset generalization.

Limitations

In general, we treat our approach as an initial framework for rule-induced cross-dataset generalization in rumor detection. In this section, we discuss three aspects that future studies can further study.

Multimodality. Our current approach only considers handling textual posts and responses. Due to the remarkable progress in multimodal LLMs, we believe that extending our framework to handle multimodal data is a promising future work.

Agentic implementation. Our framework builds upon several building block modules, which naturally fits for agentic system design, which has not been explored in the current paper. For example, the source dataset and predicate schema can be extended as dynamic databases for storing up-to-date event and predicate libraries, meanwhile the rule induction module can be integrated with external symbolic reasoning systems to further enhance robustness and consistency.

Generalization analysis. Our work provides the empirical justification that by conducting rule-based formal reasoning, the cross-dataset generalization abilities of LLMs can be significantly improved in rumor detection. While we have not studied the boundary of the generalizability for this paradigm, which can be a fundamental future research problem to explore.

Potential Broader Impact

Our research is a technical work targeting at solving the single-source cross-dataset generalization problem in rumor detection. Due to the negative influence of misinformation and rumors in real world, we believe that our approach, if properly utilized as designed, can lead to positive broader impact. We also note that for high-risk detection scenarios, our approach is not able to fully take the place of expert justification for absolute safety.

References

Jawaher Alghamdi, Yuqing Lin, and Suhui Luo. 2024. Cross-domain fake news detection using a prompt-based approach. *Future Internet*, 16(8):286.

Ahmer Arif, John J Robinson, Stephanie A Stanek, Elodie S Fichet, Paul Townsend, Zena Worku, and Kate Starbird. 2017. A closer look at the self-correcting crowd: Examining corrections in online rumors. In *Proceedings of the 2017 ACM conference*

on computer supported cooperative work and social computing, pages 155–168.

G Hanieh Asadi, Ali Hamzeh, and Niloofar Mozafari. 2025. Metadata-free rumor detection via llm-augmented reasoning. *IEEE Access*.

Tian Bian, Xi Xiao, Tingyang Xu, Peilin Zhao, Wenbing Huang, Yu Rong, and Junzhou Huang. 2020. Rumor detection on social media with bi-directional graph convolutional networks. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 549–556.

Carlos Castillo, Marcelo Mendoza, and Barbara Poblete. 2011. Information credibility on twitter. In *Proceedings of the 20th international conference on World wide web*, pages 675–684.

Hyungjoo Chae, Yeonghyeon Kim, Seungone Kim, Kai Tzu-iunn Ong, Beong-woo Kwak, Moohyeon Kim, Seonghwan Kim, Taeyoon Kwon, Jiwan Chung, Youngjae Yu, and 1 others. 2024. Language models as compilers: Simulating pseudocode execution improves algorithmic reasoning in language models. *arXiv preprint arXiv:2404.02575*.

Songlin Chen, Xiaoliang Chen, Duoqian Miao, Hongyun Zhang, Xiaolin Qin, and Peng Lu. 2025. Ada-uda: A transferable transformer framework for rumor detection using adversarial domain alignment within unsupervised domain adaptation. *Expert Systems with Applications*, 261:125487.

Yanfang Chen, Ding Chen, Shichao Song, Simin Niu, Hanyu Wang, Zeyun Tang, Feiyu Xiong, and Zhiyu Li. 2024. Hrde: Retrieval-augmented large language models for chinese health rumor detection and explainability. *arXiv preprint arXiv:2407.00668*.

Tsun-Hin Cheung and Kin-Man Lam. 2023. Factllama: Optimizing instruction-following language models with external knowledge for automated fact-checking. In *2023 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, pages 846–853. IEEE.

Chaoqun Cui and Caiyan Jia. 2024. Propagation tree is not deep: Adaptive graph contrastive learning approach for rumor detection. In *Proceedings of the AAAI Conference on artificial intelligence*, volume 38, pages 73–81.

DeepSeek-AI. 2026. Deepseek-v4: Towards highly efficient million-token context intelligence.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186.

724	Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon,	Guanghua Li, Wensheng Lu, Wei Zhang, Defu Lian,	780
725	Pengfei Liu, Yiming Yang, Jamie Callan, and Gra-	Kezhong Lu, Rui Mao, Kai Shu, and Hao Liao.	781
726	ham Neubig. 2023. Pal: Program-aided language	2024. Re-search for the truth: Multi-round retrieval-	782
727	models. In <i>International conference on machine</i>	augmented large language models are strong fake	783
728	<i>learning</i> , pages 10764–10799. PMLR.	news detectors. <i>arXiv preprint arXiv:2403.09747</i> .	784
729	Ziyu Ge, Yuhao Wu, Daniel Wai Kit Chin, Roy Ka-	Hongzhan Lin, Jing Ma, Liangliang Chen, Zhiwei Yang,	785
730	Wei Lee, and Rui Cao. 2025. Resolving con-	Mingfei Cheng, and Chen Guang. 2022. Detect ru-	786
731	flicting evidence in automated fact-checking: A	mers in microblog posts for low-resource domains	787
732	study on retrieval-augmented llms. <i>arXiv preprint</i>	via adversarial contrastive learning. In <i>Findings</i>	788
733	<i>arXiv:2505.17762</i> .	<i>of the Association for Computational Linguistics:</i>	789
734	Yuxia Gong, Shuguo Hu, and Huaiwen Zhang. 2025.	<i>NAACL 2022</i> , pages 2543–2556.	790
735	Cross-domain rumor detection via test-time adapta-	Hongzhan Lin, Pengyao Yi, Jing Ma, Haiyun Jiang,	791
736	tion and large language models. In <i>Proceedings of</i>	Ziyang Luo, Shuming Shi, and Ruifang Liu. 2023.	792
737	<i>the 2025 Conference on Empirical Methods in Natu-</i>	Zero-shot rumor detection with propagation structure	793
738	<i>ral Language Processing</i> , pages 8051–8066.	via prompt learning. In <i>Proceedings of the AAAI Con-</i>	794
739	Zejiang He, Jingyuan Huang, Menglong Lu, Zhen	<i>ference on Artificial Intelligence</i> , volume 37, pages	795
740	Huang, Shanshan Liu, Zhiliang Tian, and Dongsheng	5213–5221.	796
741	Li. 2025. Gcml: Gradient coherence guided meta-	Zhan Ling, Yunhao Fang, Xuanlin Li, Zhiao Huang,	797
742	learning for cross-domain emerging topic rumor de-	Mingu Lee, Roland Memisevic, and Hao Su. 2023.	798
743	tection. In <i>Proceedings of the 2025 Conference on</i>	Deductive verification of chain-of-thought reasoning.	799
744	<i>Empirical Methods in Natural Language Processing</i> ,	In <i>NeurIPS</i> .	800
745	pages 19148–19162.	Yifan Liu, Yaokun Liu, Zelin Li, Ruichen Yao, Yang	801
746	Joy Hsu, Emily Jin, Jiajun Wu, and Niloy J Mitra.	Zhang, and Dong Wang. 2025a. Modality interactive	802
747	2025. From programs to poses: Factored real-world	mixture-of-experts for fake news detection. In <i>Pro-</i>	803
748	scene generation via learned program libraries. <i>arXiv</i>	<i>ceedings of the ACM on Web Conference 2025</i> , pages	804
749	<i>preprint arXiv:2510.10292</i> .	5139–5150.	805
750	Beizhe Hu, Qiang Sheng, Juan Cao, Yuhui Shi, Yang	Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Man-	806
751	Li, Danding Wang, and Peng Qi. 2024. Bad actor,	dar Joshi, Danqi Chen, Omer Levy, Mike Lewis,	807
752	good advisor: Exploring the role of large language	Luke Zettlemoyer, and Veselin Stoyanov. 2019.	808
753	models in fake news detection. In <i>Proceedings of the</i>	Roberta: A robustly optimized bert pretraining ap-	809
754	<i>AAAI conference on artificial intelligence</i> , volume 38,	proach. <i>arXiv preprint arXiv:1907.11692</i> .	810
755	pages 22105–22113.	Zhiwei Liu, Kailai Yang, Qianqian Xie, Christine	811
756	Zhen Huang, Zhilong Lv, Xiaoyun Han, Binyang Li,	De Kock, Sophia Ananiadou, and Eduard Hovy.	812
757	Menglong Lu, and Dongsheng Li. 2022. Social bot-	2025b. Raemollm: Retrieval augmented llms for	813
758	aware graph neural network for early rumor detection.	cross-domain misinformation detection using in-	814
759	In <i>proceedings of the 29th international conference</i>	context learning based on emotional information. In	815
760	<i>on computational linguistics</i> , pages 6680–6690.	<i>Proceedings of the 63rd Annual Meeting of the As-</i>	816
761	Ye Jiang, Taihang Wang, Xiaoman Xu, Yimin Wang,	<i>sociation for Computational Linguistics (Volume 1:</i>	817
762	Xingyi Song, and Diana Maynard. 2025. Cross-	<i>Long Papers)</i> , pages 16508–16523.	818
763	modal augmentation for few-shot multimodal fake	Jing Ma, Wei Gao, Prasenjit Mitra, Sejeong Kwon,	819
764	news detection. <i>Engineering Applications of Artifi-</i>	Bernard J Jansen, Kam-Fai Wong, and Meeyoung	820
765	<i>cial Intelligence</i> , 142:109931.	Cha. 2016. Detecting rumors from microblogs with	821
766	R Kenny Jones, Paul Guerrero, Niloy J Mitra, and	recurrent neural networks.	822
767	Daniel Ritchie. 2025. Shapelib: Designing a library	Jing Ma, Wei Gao, Zhongyu Wei, Yueming Lu, and	823
768	of programmatic 3d shape abstractions with large	Kam-Fai Wong. 2015. Detect rumors using time se-	824
769	language models. <i>arXiv preprint arXiv:2502.08884</i> .	ries of social context information on microblogging	825
770	Elena Kochkina, Maria Liakata, and Arkaitz Zubiaga.	websites. In <i>Proceedings of the 24th ACM interna-</i>	826
771	2018. All-in-one: Multi-task learning for rumour	<i>tional on conference on information and knowledge</i>	827
772	verification. In <i>Proceedings of the 27th international</i>	<i>management</i> , pages 1751–1754.	828
773	<i>conference on computational linguistics</i> , pages 3402–	Jing Ma, Wei Gao, and Kam-Fai Wong. 2017. Detect	829
774	3413.	rumors in microblog posts using propagation struc-	830
775	Chengshu Li, Jacky Liang, Andy Zeng, Xinyun Chen,	ture via kernel learning. In <i>Proceedings of the 55th</i>	831
776	Karol Hausman, Dorsa Sadigh, Sergey Levine, Li Fei-	<i>Annual Meeting of the Association for Computational</i>	832
777	Fei, Fei Xia, and Brian Ichter. 2023. Chain of code:	<i>Linguistics (Volume 1: Long Papers)</i> , pages 708–717.	833
778	Reasoning with a language model-augmented code	Jing Ma, Wei Gao, and Kam-Fai Wong. 2018a. De-	834
779	emulator. <i>arXiv preprint arXiv:2312.04474</i> .	tect rumor and stance jointly by neural multi-task	835

836	learning. In <i>Companion proceedings of the the web conference 2018</i> , pages 585–593.	
837		
838	Jing Ma, Wei Gao, and Kam-Fai Wong. 2018b. Rumor	
839	detection on twitter with tree-structured recursive	
840	neural networks. In <i>Proceedings of the 56th Annual</i>	
841	<i>Meeting of the Association for Computational Lin-</i>	
842	<i>guistics (Volume 1: Long Papers)</i> , pages 1980–1989.	
843	Jing Ma, Jun Li, Wei Gao, Yang Yang, and Kam-Fai	
844	Wong. 2021. Improving rumor detection by promot-	
845	ing information campaigns with transformer-based	
846	generative adversarial learning. <i>IEEE Transactions</i>	
847	<i>on Knowledge and Data Engineering</i> , 35(3):2657–	
848	2670.	
849	Qiong Nan, Juan Cao, Yongchun Zhu, Yanyan Wang,	
850	and Jintao Li. 2021. Mdfend: Multi-domain fake	
851	news detection. In <i>Proceedings of the 30th ACM in-</i>	
852	<i>ternational conference on information & knowledge</i>	
853	<i>management</i> , pages 3343–3347.	
854	OpenAI. 2024. Hello GPT-4o .	
855	Yifei Peng, Yaoli Liu, Enbo Xia, Yu Jin, Wang-Zhou	
856	Dai, Zhong Ren, Yao-Xiang Ding, and Kun Zhou.	
857	2025. Abductive logical rule induction by bridging	
858	inductive logic programming and multimodal large	
859	language models. <i>arXiv preprint arXiv:2509.21874</i> .	
860	Wasu Top Piriyakulkij, Yichao Liang, Hao Tang,	
861	Adrian Weller, Marta Kryven, and Kevin Ellis. 2025.	
862	Poe-world: Compositional world modeling with	
863	products of programmatic experts. <i>arXiv preprint</i>	
864	<i>arXiv:2505.10819</i> .	
865	Reid Pryzant, Ziyi Yang, Yichong Xu, Chenguang Zhu,	
866	and Michael Zeng. 2022. Automatic rule induction	
867	for efficient semi-supervised learning. In <i>Findings</i>	
868	<i>of the Association for Computational Linguistics:</i>	
869	<i>EMNLP 2022</i> , pages 28–44.	
870	Vahed Qazvinian, Emily Rosengren, Dragomir Radev,	
871	and Qiaozhu Mei. 2011. Rumor has it: Identifying	
872	misinformation in microblogs. In <i>Proceedings of</i>	
873	<i>the 2011 conference on empirical methods in natural</i>	
874	<i>language processing</i> , pages 1589–1599.	
875	Litao Qiao, Weijia Wang, and Bill Lin. 2021. Learn-	
876	ing accurate and interpretable decision rule sets from	
877	neural networks. In <i>Proceedings of the AAAI con-</i>	
878	<i>ference on artificial intelligence</i> , volume 35, pages	
879	4303–4311.	
880	Zhen Qin, Yicheng Cheng, Zhe Zhao, Zhe Chen, Donald	
881	Metzler, and Jingzheng Qin. 2020. Multitask mixture	
882	of sequential experts for user activity streams. In	
883	<i>Proceedings of the 26th ACM SIGKDD international</i>	
884	<i>conference on knowledge discovery & data mining</i> ,	
885	pages 3083–3091.	
886	Qwen Team. 2026. Qwen3.5: Towards native multi-	
887	modal agents .	
	Hongyan Ran and Caiyan Jia. 2023. Unsupervised	888
	cross-domain rumor detection with contrastive learn-	889
	ing and cross-attention. In <i>Proceedings of the aaai</i>	890
	<i>conference on artificial intelligence</i> , volume 37,	891
	pages 13510–13518.	892
	Hongyan Ran, Di Zhang, Xiaohong Li, Huifang Ma,	893
	Caiyan Jia, and Yaogong Feng. 2026. Unsupervised	894
	contrastive domain adaptive rumor detection with	895
	test-time classifier adjustment. <i>Information Process-</i>	896
	<i>ing & Management</i> , 63(2):104341.	897
	Kai Shu, Limeng Cui, Suhang Wang, Dongwon Lee,	898
	and Huan Liu. 2019. defend: Explainable fake news	899
	detection. In <i>Proceedings of the 25th ACM SIGKDD</i>	900
	<i>international conference on knowledge discovery &</i>	901
	<i>data mining</i> , pages 395–405.	902
	Amila Silva, Ling Luo, Shanika Karunasekera, and	903
	Christopher Leckie. 2021. Embracing domain differ-	904
	ences in fake news: Cross-domain fake news detec-	905
	tion using multi-modal data. In <i>Proceedings of the</i>	906
	<i>AAAI conference on artificial intelligence</i> , volume 35,	907
	pages 557–565.	908
	Ronit Singal, Pransh Patwa, Parth Patwa, Aman Chadha,	909
	and Amitava Das. 2024. Evidence-backed fact check-	910
	ing using rag and few-shot in-context learning with	911
	llms. In <i>Proceedings of the Seventh Fact Extraction</i>	912
	<i>and VERification Workshop (FEVER)</i> , pages 91–98.	913
	Kate Starbird, Ahmer Arif, Tom Wilson, Katherine	914
	Van Koeveering, Katya Yefimova, and Daniel Scarnec-	915
	chia. 2018. Ecosystem or echo-system? exploring	916
	content sharing across alternative media domains. In	917
	<i>Proceedings of the International AAAI Conference</i>	918
	<i>on Web and Social Media</i> , volume 12.	919
	Mengzhu Sun, Xi Zhang, Jiaqi Zheng, and Guixiang	920
	Ma. 2022. Ddgc: Dual dynamic graph convolu-	921
	tional networks for rumor detection on social media.	922
	In <i>Proceedings of the AAAI conference on artificial</i>	923
	<i>intelligence</i> , volume 36, pages 4611–4619.	924
	Zhiliang Tian, Jingyuan Huang, Zejiang He, Zhen	925
	Huang, Menglong Lu, Linbo Qiao, Songzhu Mei,	926
	Yijie Wang, and Dongsheng Li. 2025. Llm-based ru-	927
	mor detection via influence guided sample selection	928
	and game-based perspective analysis. In <i>Proceed-</i>	929
	<i>ings of the 63rd Annual Meeting of the Association</i>	930
	<i>for Computational Linguistics (Volume 1: Long Pa-</i>	931
	<i>pers)</i> , pages 28402–28414.	932
	Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten	933
	Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou,	934
	and 1 others. 2022. Chain-of-thought prompting elic-	935
	its reasoning in large language models. <i>Advances</i>	936
	<i>in neural information processing systems</i> , 35:24824–	937
	24837.	938
	Zhenrui Yue, Huimin Zeng, Yang Zhang, Lanyu Shang,	939
	and Dong Wang. 2023. Metaadapt: Domain adaptive	940
	few-shot misinformation detection via meta learning.	941
	In <i>Proceedings of the 61st Annual Meeting of the</i>	942
	<i>Association for Computational Linguistics (Volume</i>	943
	<i>1: Long Papers)</i> , pages 5223–5239.	944

Fengzhu Zeng, Qian Shao, Ling Cheng, Wei Gao, Shih-Fen Cheng, Jing Ma, and Cheng Niu. 2026. Llm-based few-shot early rumor detection with imitation agent. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*, pages 1856–1867.

Zhe Zhao, Paul Resnick, and Qiaozhu Mei. 2015. Enquiring minds: Early detection of rumors in social media from enquiry posts. In *Proceedings of the 24th international conference on world wide web*, pages 1395–1405.

Zhaocheng Zhu, Yuan Xue, Xinyun Chen, Denny Zhou, Jian Tang, Dale Schuurmans, and Hanjun Dai. 2023. Large language models can learn rules. *arXiv preprint arXiv:2310.07064*.

Arkaitz Zubiaga, Ahmet Aker, Kalina Bontcheva, Maria Liakata, and Rob Procter. 2018. Detection and resolution of rumours in social media: A survey. *Acm Computing Surveys (Csur)*, 51(2):1–36.

A Predicates

The semantic predicate schema used in our framework is summarized in Table 4. These predicates are designed to cover general rumor-related cues, including source credibility, evidential support, claim specificity, uncertainty, communicative intent, and user reactions. We select these cues because they are relatively independent of specific topics or events and can provide transferable evidence for cross-domain rumor detection. Keeping the schema compact also helps maintain interpretability and reduces noise during rule induction.

B Rule Induction Analysis

Table 5 presents a representative rulebook learned from the GPT-4o-based configuration under the PHEME→Twitter16 transfer scenario, which is one of the better-performing settings in our experiments. We first identify reliable source instances based on the predictions of the pretrained GCN backbone, retrieve their neighboring examples, and abstract their textual and conversational signals into Boolean predicates. Candidate rules are then constructed as single predicates or short conjunctions, and the most class-discriminative ones are retained in the final rulebook.

The induced rules reveal several interpretable rumor-indicative patterns. For example, they highlight claims from non-institutional sources, speculative or accusatory wording, missing or vague evidence, and skeptical reactions in early replies. These cues are useful because early rumor detection often has limited contextual evidence, making

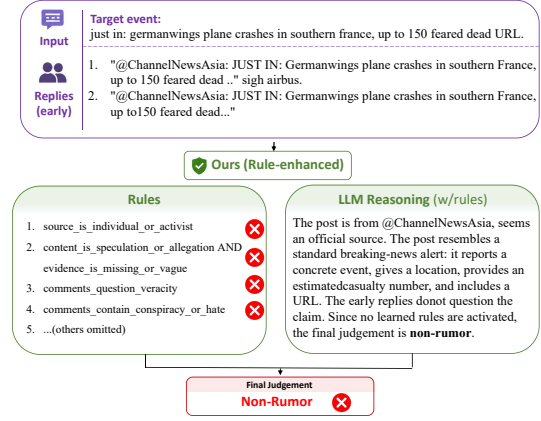


Figure 5: A failure case of RIFT.

the LLM’s zero-shot judgment unstable. By injecting the learned rules as soft evidence, our method makes such source-domain regularities explicit and guides the LLM to consider transferable discriminative signals during target-domain inference.

ID	Rulebook
R1	source_is_individual_or_activist
R2	content_is_speculation_or_allegation AND evidence_is_missing_or_vague
R3	comments_question_veracity
R4	evidence_is_missing_or_vague AND comments_question_veracity
R5	comments_contain_conspiracy_or_hate

Table 5: Representative rules induced from the source domain.

C Failure Case

Figure 5 shows a failure case where our method incorrectly predicts a rumor-labeled event as non-rumor. The post resembles a credible breaking-news alert, with a concrete event, location, casualty estimate, and URL. Moreover, the early replies merely repeat the report or express emotional reactions, rather than questioning its veracity or providing corrective context. Consequently, no rumor-side rule is activated, and the LLM produces a non-rumor judgment. This case suggests that rule-enhanced inference can be less effective when the early context provides few discriminative cues, especially in breaking-news scenarios where posts resemble credible news alerts and replies mainly echo the report.

Predicate	Description
source_is_official_media	The author is an official media, institutional, government, or law-enforcement account.
source_is_individual_or_activist	The author is an individual user, activist account, organization account, or otherwise unverified source.
content_is_breaking_news_update	The post reports a neutral breaking-news update.
content_is_speculation_or_allegation	The post contains unsupported speculation, allegation, or conspiratorial framing.
intent_is_opinion_or_solidarity	The post primarily expresses subjective opinions or emotions rather than making a verifiable factual claim.
evidence_is_missing_or_vague	The post lacks concrete evidence, links, images, or relies.
comments_question_veracity	Replies express skepticism toward the claim or its source.
comments_contain_conspiracy_or_hate	Replies contain conspiracy claims, hateful language, or hostile framing.

Table 4: Semantic predicates used for predicate grounding.

Statistic	PHEME	Twitter16	Twitter-COVID19
Events	6,425	818	400
Tree nodes	105,354	19,376	406,185
Non-rumors	4,023	205	148
Rumors	2,402	613	252
Avg. # posts/tree	16	24	1,015

Table 6: Statistics of the datasets.

D Implementation Details

We adopted BiGCN and DDGCN as the pretrained GCN backbones for obtaining source-dataset representations and predictions. The statistics of the datasets used in our experiments are summarized in Table 6. The backbone parameter settings are as follows.

For DDGCN, we used the full dual-graph configuration with both propagation and knowledge graph branches. The graph embedding size and hidden size were set to 128, with three graph convolution layers. Models were trained with Adam using a learning rate of 5×10^{-5} , batch size 64, and early stopping patience of 10. For BiGCN, we used both top-down and bottom-up propagation GCN branches. The input feature dimension was set to 5000, and both hidden and output dimensions were set to 64. Models were trained with Adam using a learning rate of 5×10^{-4} , weight decay of 10^{-4} , batch size 128, and a maximum of 200 epochs. The edge dropout rates for both top-down and bottom-up branches were set to 0.2, and early stopping patience was set to 10. All experiments are executed on a computation server with NVIDIA RTX 4090 GPUs.

E Prompt Templates

We use LLM prompts for both LLM grounding module and LLM inference module. In the grounding stage, the LLM maps each target or neighbor post to Boolean predicate activations according to a predefined predicate schema, as shown in Table 7. In the inference stage, the LLM predicts whether the target post is a rumor based on the post and early replies. Tables 8 and 9 present the zero-shot and rule-enhanced reasoning prompts, respectively. Compared with the zero-shot version, the rule-enhanced prompt only adds a transferred rule module, where induced source-domain rules are provided for the final judgment.

F LLM Usage

In our research process, LLMs play no significant roles in idea formulation, method design and experimental evaluation. While they are utilized as an important tool in paper polishing.

Prompt for LLM Grounding

System Prompt

You are an expert Data Annotator for social media content analysis.
Your task is to evaluate each tweet, including the TARGET and all retrieved NEIGHBORS, against a list of predicates.
You must evaluate each tweet independently based only on the provided text.

Input Data

I have a list of tweets. One is the TARGET, and the others are retrieved NEIGHBORS.
For each tweet, I provide the [Source Text] and [Early Comments].

{FORMATTED_DATA}

Feature Schema

Please evaluate each tweet against the following {N} predicates.
Output "true" only if the tweet strongly exhibits the characteristic; otherwise output "false".

1. {PREDICATE_NAME_1}:
 - {PREDICATE_DESCRIPTION_1}
- ...
- {N}. {PREDICATE_NAME_N}:
 - {PREDICATE_DESCRIPTION_N}

Output Format

Return a valid JSON object where keys are the tweet IDs, such as "target", "n1", and "n2", and values are boolean feature maps.

```
{
  "target": {
    "{PREDICATE_NAME_1}": true,
    "{PREDICATE_NAME_2}": false
  },
  "n1": {
    "{PREDICATE_NAME_1}": false,
    "{PREDICATE_NAME_2}": true
  }
}
```

Important Notes

- Evaluate only based on the source text and comments provided.
- Evaluate each tweet independently.
- Be objective and consistent.
- Return only the JSON object.
- Do not provide additional explanation or Markdown code blocks.

Table 7: Prompt template for LLM grounding.

Prompt for Zero-shot LLM Reasoning

```
### System Prompt
You are a social media rumor detection expert.
Your task is to analyze a target post and its early replies, and determine whether the target
    post is a rumor or non-rumor.

Base your judgment on the target post as the primary object of classification.
Use the early replies only as contextual signals that may help interpret how the post was
    understood or assessed at the time.

### User Prompt
Please analyze the following post to determine whether it is a rumor or not.

### Post to Analyze
{TEST_TEXT}

### Early Replies
1. {REPLY_TEXT_1}
...
{M}. {REPLY_TEXT_M}

### Task Definition
In this task, a "Rumor" is a social media post that presents a concrete factual claim whose
    truth is uncertain, unverified, disputed, speculative, or not sufficiently supported by
    the available context at the time of posting.

A "Non-rumor" is a post whose main content is adequately supported, not substantially disputed
    , or mainly expresses opinion, emotion, humor, or personal reaction.

When the evidence is mixed, choose the label that best reflects the overall status of the main
    claim at the time of posting.

### Analysis
Analyze the target post and early replies according to the task definition.

### Final Judgment
<Rumor> or <Non-rumor>
```

Table 8: Prompt template for zero-shot LLM reasoning.

Prompt for Rule-enhanced LLM Reasoning

```
### System Prompt
You are a social media rumor detection expert.
Your task is to analyze a target post and its early replies, and determine whether the target
    post is a rumor or non-rumor.

Base your judgment on the target post as the primary object of classification.
Use the early replies only as contextual signals that may help interpret how the post was
    understood or assessed at the time.

You will also receive a small rule book transferred from another dataset, together with
    explanations of the predicates used in those rules.
Use the transferred rule book as soft evidence rather than rigid logic.
If a transferred rule appears mismatched to the current post, say so and down-weight it.

### User Prompt
Please analyze the following post to determine whether it is a rumor or not.

### Post to Analyze
{TEST_TEXT}

### Early Replies
1. {REPLY_TEXT_1}
...
{M}. {REPLY_TEXT_M}

### Task Definition
In this task, a "Rumor" is a social media post that presents a concrete factual claim whose
    truth is uncertain, unverified, disputed, speculative, or not sufficiently supported by
    the available context at the time of posting.

A "Non-rumor" is a post whose main content is adequately supported, not substantially disputed
    , or mainly expresses opinion, emotion, humor, or personal reaction.

When the evidence is mixed, choose the label that best reflects the overall status of the main
    claim at the time of posting.

### Transferred Rule Module
The following rules are transferred from source-domain high-confidence samples.
Use them as auxiliary evidence only.
Do not treat them as hard constraints.
If a rule appears mismatched to the current post, explicitly say so and down-weight it.

#### Rumor Heuristic Rules
1. {RUMOR_RULE_1}
...
{K}. {RUMOR_RULE_K}

#### Predicate Descriptions
1. {PREDICATE_NAME_1}: {PREDICATE_DESCRIPTION_1}
...
{N}. {PREDICATE_NAME_N}: {PREDICATE_DESCRIPTION_N}

### Analysis
Analyze the target post and early replies according to the task definition.
Then discuss which transferred rules, if any, are applicable or mismatched.

### Rule Reference
<which transferred rules support Rumor, which do not fit, and why>

### Applied Rules
- Rumor-side: <comma-separated rule names or None>

### Final Judgment
<Rumor> or <Non-rumor>
```

Table 9: Prompt template for rule-enhanced LLM reasoning.